

Mingen Huang

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Photoshop

AI

Word

Excel

Hobbies

Movie

Basketball

Music

Bike

Hike

Read

EDUCATION

Master's Degree Candidate in Artificial Intelligence Sep 2

Shantou University, School of Mathematics and Computer Science, China

Selected coursework: Big Data Management and Mining, Formal Methods, Modern Database Theory,Advanced Software Engineering, Algorithms and Intelligence, Combinatorics, Software Architecture and Design Patterns

Bachelor's Degree in Information and Computing Science S

Suzhou University of Science and Technology, China

Selected coursework: Data Structures, Mathematical Analysis, Advanced Algebra, Python, C++, C, and Java

Research interests

Complex networks; Urban mobility and transportation networks; Human mobility; Data analysis; Graph-based modelling; Deep learning; Date mining

RESEARCH EXPERIENCE

Extreme Climate Modelling and Meteorological Data Analysis: Suzhou

- Taking the time series of annual mean temperatures in the Suzhou area from 1951 to 2021 as the subject of study, this research analyzes climate change trends and abrupt change characteristics using methods such as the M-K abrupt change test, the BFAST test, and linear regression forecasting. The results indicate that the annual mean temperature in Suzhou will continue to rise over the next five years, and the frequency of extreme weather events may increase.

Spatiotemporal characterization and contextual interpretation of taxi route deviations using GPS trajectories

Manuscript submitted

- To address the limitations of traditional methods that rely solely on path length to identify abnormal taxi detours, this study proposes a two-dimensional RDI-TDI analysis framework that integrates the degree of relative spatial detour and the degree of temporal travel inefficiency. Based on GPS trajectory, road network, and POI datas from taxis in Chengdu, a two-dimensional Gaussian Mixture Model is employed to classify taxi trips into four categories: normal trips, congestion trips, benign detours, and extreme spatio-temporal inefficient trips.

MODELLING EXPERIENCE

21st China Postgraduate Mathematical Contest in Modelling (Huawei Cup)

Second Prize

- This study analyzes video data from a5-km highway segment with two traffic lanes and an emergency lane. YOLOv5 and DeepSORT are used to extract traffic parameters, which support a congestion index model integrating regression and neural-network methods to provide warnings about persistent congestion approximately10 minutes in advance. A Weibull-based model guides emergency-lane activation, while entropy-weighted TOPSIS evaluates its impact on congestion and traffic efficiency. Finally, LSTM prediction and Pareto-based simulated annealing are combined to optimize monitoring-point placement, balancing monitoring performance, cost, and installation count.

13th Teddy Cup Data Mining Challenge

First Prize

- A comprehensive paper-screening system was developed to analyze structural, semantic, and multimodal content. PyMuPDF and pdfplumber extracted key text elements and basic statistics, while PDF repair, OCR, and large language models processed scanned documents. Fuzzy matching detected potential team-information leaks, and TF-IDF–based cosine similarity assessed topic relevance. Random Forest and XGBoost were combined to exclude papers lacking substantive content. For plagiarism detection, TF-IDF, Vision Transformers, and MD5/structural similarity were used to compare text, images, and formulas. A dynamic weighting strategy further improved similarity assessment for image-intensive papers.